# Experiment No. 8

## Aim: Implementation of a program for designing a GUI using Swing components.

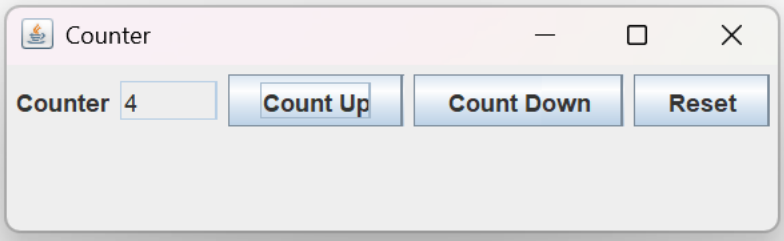
1. Implement the following GUI without any IDE

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class CounterGUI extends JFrame implements ActionListener {

    JTextField textField;
    JButton btnUp, btnDown, btnReset;
    int count = 0;

    public CounterGUI() {
        setTitle("Counter");
        setSize(400, 120);
        setLayout(new FlowLayout());

        JLabel label = new JLabel("Counter");

        textField = new JTextField("0", 5);
        textField.setEditable(false);

        btnUp = new JButton("Count Up");
        btnDown = new JButton("Count Down");
        btnReset = new JButton("Reset");

        add(label);
        add(textField);
        add(btnUp);
        add(btnDown);
        add(btnReset);

        btnUp.addActionListener(this);
        btnDown.addActionListener(this);
        btnReset.addActionListener(this);
```

```
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setVisible(true);
    }

    public void actionPerformed(ActionEvent e) {
        if (e.getSource() == btnUp) {
            count++;
        } else if (e.getSource() == btnDown) {
            count--;
        } else if (e.getSource() == btnReset) {
            count = 0;
        }

        textField.setText(String.valueOf(count));
    }

    public static void main(String[] args) {
        new CounterGUI();
    }
}
```

OUTPUT:

```
c:\Users\PRASANNA\Desktop\AOOC-Practicals-4th sem\exp8>cd "c:\Users\PRASANNA\Des
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
```



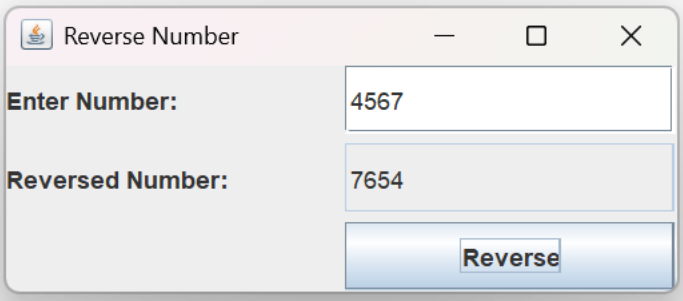
2. Write a GUI program to find the reverse of a given number using Swing (with IDE)

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class ReverseNumberGUI extends JFrame implements ActionListener {

    JTextField inputField, resultField;
    JButton reverseBtn;

    public ReverseNumberGUI() {
        setTitle("Reverse Number");
        setSize(350, 150);
        setLayout(new GridLayout(3, 2, 5, 5));

        JLabel inputLabel = new JLabel("Enter Number:");
        JLabel resultLabel = new JLabel("Reversed Number:");

        inputField = new JTextField();
        resultField = new JTextField();
        resultField.setEditable(false);

        reverseBtn = new JButton("Reverse");

        add(inputLabel);
        add(inputField);
        add(resultLabel);
        add(resultField);
        add(new JLabel()); // empty space
        add(reverseBtn);

        reverseBtn.addActionListener(this);

        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setVisible(true);
    }

    public void actionPerformed(ActionEvent e) {
        try {
            int num = Integer.parseInt(inputField.getText());
            int reversed = 0;

            while (num != 0) {
```

```
            int digit = num % 10;
            reversed = reversed * 10 + digit;
            num = num / 10;
        }

        resultField.setText(String.valueOf(reversed));

    } catch (NumberFormatException ex) {
        JOptionPane.showMessageDialog(this, "Please enter a valid number!");
    }
}

public static void main(String[] args) {
    new ReverseNumberGUI();
}
}
```

OUTPUT:

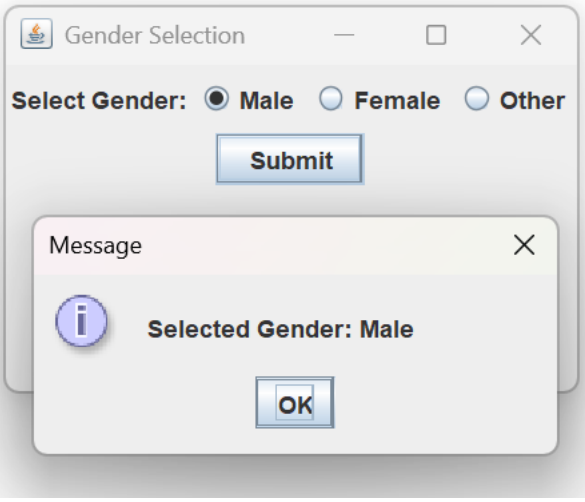
```
Microsoft Windows [Version 10.0.26200.8117]
(c) Microsoft Corporation. All rights reserved.

C:\Users\PRASANNA\Desktop\AOOC-Practicals-4th sem\exp8> cmd /C ""C:\Program File
ppData\Roaming\Code\User\workspaceStorage\b12d66b0b55c3bd7693a4f80be7edd25\redha
```



3. Write a GUI program to demonstrate the use of radio buttons (e.g., gender selection)

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class GenderGUI extends JFrame implements ActionListener {

    JRadioButton male, female, other;
    JButton submit;

    public GenderGUI() {
        setTitle("Gender Selection");
        setSize(300, 200);
        setLayout(new FlowLayout());

        JLabel label = new JLabel("Select Gender:");

        // Create radio buttons
        male = new JRadioButton("Male");
        female = new JRadioButton("Female");
        other = new JRadioButton("Other");

        // Group them
        ButtonGroup bg = new ButtonGroup();
        bg.add(male);
        bg.add(female);
        bg.add(other);

        // Button
        submit = new JButton("Submit");

        // Add components
        add(label);
        add(male);
        add(female);
        add(other);
        add(submit);

        // Action
        submit.addActionListener(this);

        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setVisible(true);
    }
```

```
    public void actionPerformed(ActionEvent e) {
        String gender;

        if (male.isSelected()) {
            gender = "Male";
        } else if (female.isSelected()) {
            gender = "Female";
        } else if (other.isSelected()) {
            gender = "Other";
        } else {
            gender = "Not Selected";
        }

        JOptionPane.showMessageDialog(this, "Selected Gender: " + gender);
    }

    public static void main(String[] args) {
        new GenderGUI();
    }
}
```

OUTPUT :

```
c:\Users\PRASANNA\Desktop\AOOC-Practicals-4th sem\exp8>cd "c:\Users\PRASANNA\
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
```